

Reducing Customer Churn

The carrier loses **26.5% of customers a year** — about **\$1.67M** in annual recurring revenue. A model-targeted retention play can net an estimated **~\$347K/year at 2.7x ROI**.

The question

Who is most likely to leave, why, and is it worth paying to keep them?

What we found

1. It's expensive churn. \$1.67M/yr walks out the door, and churners pay **more** than those who stay (\$74 vs \$61/mo) — so retention pays.

2. The drivers are clear and consistent. Month-to-month contracts churn **43%** vs **3%** on two-year; **53%** of new customers leave in the first six months; electronic-check (45%) and no-tech-support (42%) customers are high-risk.

3. It's predictable. A logistic-regression model (test **AUC 0.85**) puts **67% of all churn in the top 30%** of customers by risk — the basis for efficient targeting.

Recommendation

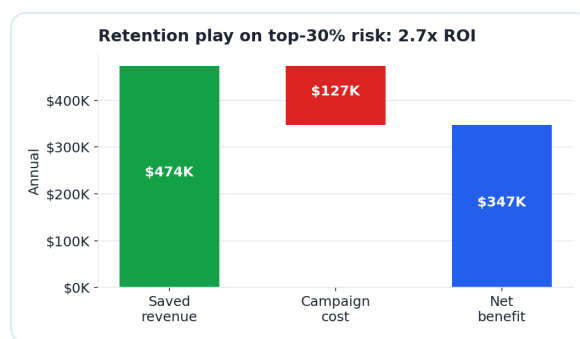
- 1. Aim, don't spray.** Focus retention on the top 30% risk (2x budget efficiency).
- 2. Attack contract type.** Move high-risk month-to-month customers onto 1–2 year terms.
- 3. Fix the first six months.** Invest in onboarding and bundle tech support / online security (both roughly halve churn).
- 4. Shift electronic-check payers to autopay** — the highest-churn payment method.

\$347K

estimated net benefit / year

2.7x

return on retention spend



Base case: a \$60 offer to the top-30% risk segment (2,113 customers) at a 30% save rate. Positive down to a 15% save rate.

Data: IBM Telco Customer Churn (7,043 customers, public). **Method:** pandas EDA + logistic-regression churn model (scikit-learn), risk-decile targeting, scenario ROI. **Next step:** validate the save-rate assumption with a small A/B holdout before scaling.